

# ABHAY RAVI KUMAR

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## TECHNICAL SKILLS

**Programming Languages:** Python, Java, C++, Kotlin, SQL, JavaScript, Scala, TypeScript.

**Frameworks & Libraries:** Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Flask, React.js, Node.js.

**Big Data & Cloud Technologies:** Apache Spark, Hadoop, Hive, Kafka, AWS (EMR, S3, Lambda), Docker, Kubernetes.

**Data Analytics & Visualization:** Power BI, Tableau, Qlik, Excel, Matplotlib, Seaborn, Plotly, Dash.

**Software Development:** Object-Oriented Programming (OOP), Design Patterns, REST APIs, Git, Agile/Scrum.

**Databases:** MySQL, PostgreSQL, MongoDB, Room (Android), SQLite.

**Machine Learning & NLP:** Regression, Classification, Clustering, Natural Language Processing, Transformers, Generative AI, Large language models(LLM's).

## WORK EXPERIENCE

### Procurement Operations Analyst

Feb 2023 – Dec 2023

Hewlett Packard Enterprise

Bengaluru, India

- Analyzed global procurement and inventory data from SAP to identify bottlenecks in CPU and HDD shipments, leading to a **12% improvement** in average delivery time across APAC.
- Built real-time dashboards using Power BI and Qlik to monitor stock levels, vendor performance, and shipment delays—adopted by regional leads and planners, reducing stockout-related escalations by **30%**.
- Performed root-cause analysis on delayed orders and inventory buildup using Python and time-series data, contributing to a **15% reduction** in inventory holding costs over three quarters.
- Collaborated with supply chain engineers and planners to translate operational pain points into data models and KPIs, streamlining replenishment forecasting for **250+ parts**.

### Intern

Aug 2022 – Sept 2022

Tata Elxsi

Bengaluru, India

- Redesigned app architecture using MVVM, Repository pattern, Room, and LiveData—cutting screen load time by **30%** and reducing UI response latency by **150ms**.
- Optimized background task execution with Coroutines and WorkManager—reduced frame drops by **25%** and improved workflow stability across **5+ UI components**.
- Built a media playback module using ExoPlayer, MVVM, and HILT—achieved smooth streaming with **40% fewer** buffering events under weak networks via adaptive bitrate handling.

## PROJECTS

### Resume Parser: Leveraging NLP for Candidate Profiling | Python, SpaCy, Tesseract OCR, Flask, Regex

- Built a resume parsing engine using NLP and OCR to extract structured data from PDF and scanned documents.
- Streamlined resume screening for large datasets, enabling fast and accurate candidate profiling.
- Reached **92% precision** in entity extraction; reduced resume screening time by **80%**.
- Followed a modular and layered approach to ensure scalability, efficiency, and real-time processing for recruitment platforms, utilizing Python for advanced text processing and entity recognition.

### Supply Chain Optimizer | Python, OR-Tools, Pandas, Dash, Plotly

- Developed a logistics optimization tool using OR-Tools to simulate and optimize multi-warehouse inventory distribution and delivery routing.
- Enabled data-driven decisions for stock allocation and reduced simulated delivery time and stockouts in test scenarios.
- Achieved **25%** improvement in delivery efficiency and **40%** reduction in stockouts across simulated models.

### Big Data Pipeline for IoT Device Monitoring | Scala, Apache Spark, AWS EMR/S3, JUnit, Python

- Built a scalable Spark pipeline on **AWS EMR** to process 2 TB/day of IoT sensor data in real time and batch modes.
- Optimized performance by tuning Spark configurations and job parallelism, reducing run time by **35%**.
- Implemented schema-aware ingestion with Avro/Parquet and validated ETL workflows using **JUnit**.

## EDUCATION

### Stony Brook University

Jan 2024 – Dec 2025

Master of Science in Computer Science

Stony Brook, NY

### RV Institute of Technology and Management

Aug 2019 – May 2023

Bachelor of Technology in Information Science Engineering

Bengaluru, India